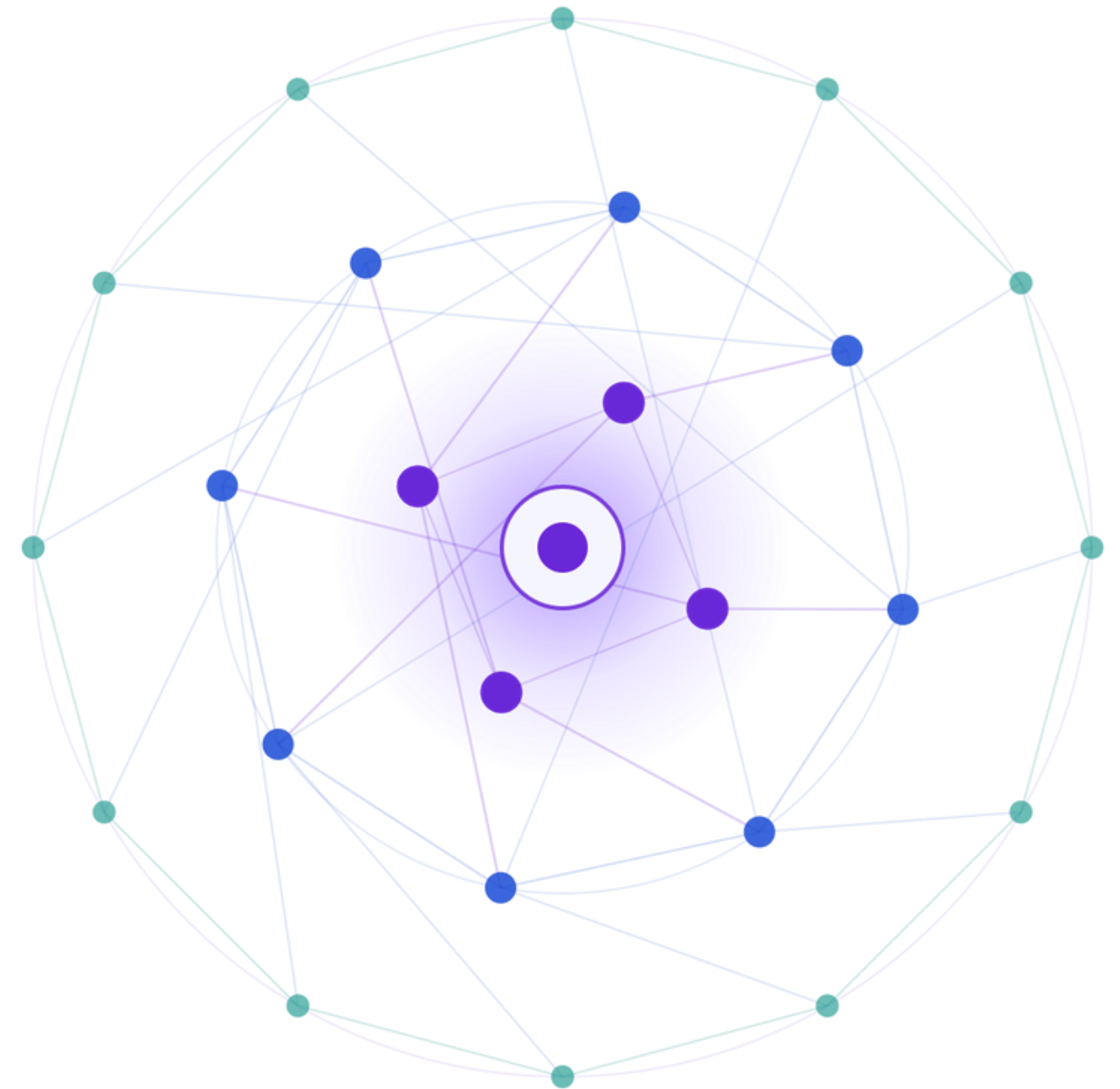


SAMSUNG PRISM

# Generative AI Hackathon

**3rd Edition** 2026 – 27

- Theme ID - 2
- Team Name - The Cyclops
- College Name - SRM Institute of Science and Technology
- Member Name & Email 1- Nishant Ranjan & nishant.ranjan.air1@gmail.com
- Member Name & Email 2- G. Vishal & valiantvishal30@gmail.com
- Member Name & Email 3- Nidhi Nayana & nidhinayana0412@gmail.com
- Member Name & Email 4- Rangesh Pandian & rangeshpandian@gmail.com
- Submission Github link - [https://github.com/Nishant-codess/fixby\\_galaxy\\_engine](https://github.com/Nishant-codess/fixby_galaxy_engine)



# Theme

*575+ Settings. One Frustrated User. Zero Tolerance for Wrong Answers.*

Samsung Galaxy devices have 575+ nested settings screens. Real users don't say "adjust background usage limits." They say:

"bhai mera phone bohot garam ho raha hai aur battery jaldi khatam ho rahi hai"

The challenge: Turn messy, multilingual complaints into precise, safe, one-tap troubleshooting plans — with verified Samsung One UI deeplinks, not hallucinated web URLs.

3 things that go wrong today:

## 1. Hallucinated URLs

LLMs invent dead links like [samsung.com/support/battery](https://samsung.com/support/battery) instead of real `bixby://` deeplinks

## 2. Wrong menu depth

"Go to Settings > Battery" leaves the user stranded. The fix is 4 levels deep.

## 3. Unsafe ordering

Factory Reset suggested as Step 1 instead of safe adjustments first

# Existing Solutions & Gaps

*Why a Generic LLM Can't Solve This*

A normal chatbot gives you a paragraph. Theme 02 needs Samsung-exact structured JSON.

## What generic LLMs do wrong:

- Output free-text instead of strict JSON schema
- Hallucinate HTTP URLs instead of bixby:// deeplinks
- Land on root menus instead of the deepest leaf screen
- Suggest Factory Reset before safe options
- Take 2,000-4,000 ms per response
- Fail on Hinglish and Korean input

## What Theme 02 actually requires:

- Strict schema: Title 2-3 words, Description 5-7 words starting with "It will"
- Only verified bixby://settings/ deeplinks
- Deepest leaf screen resolution
- Non-invasive steps first, critical steps last
- Under 300 ms for cached queries
- English + Hinglish + Korean support

**Key insight:** *You can't prompt-engineer your way to all of these. You need an engineered system around the model.*

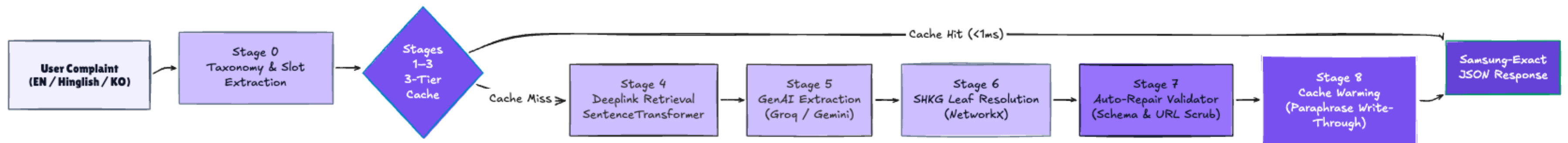
# Our Solutions & Architecture Diagram

*Fixby: 8-Stage Deterministic Pipeline*

GenAI is used in exactly 1 of 8 stages. Everything else is deterministic code.

## The 8 Stages:

- Stage 0: Symptom Taxonomy & Slot Extraction — classifies domain + symptom from any language (< 0.1 ms)
- Stages 1-3: 3-Tier Semantic Cache — Tier 1: MD5 hash, Tier 2: domain+symptom slot hash, Tier 3: MiniLM vector cosine (< 1 ms on hit)
- Stage 4: Candidate Deeplink Retrieval — SentenceTransformer matches query against 112 Samsung deeplinks (< 50 ms)
- Stage 5: GenAI Schema Extraction — LLM (Groq or Gemini) generates structured plan from bounded candidates (~200 ms) ← ONLY GenAI STAGE
- Stage 6: SHKG Leaf Resolution — NetworkX graph traverses to deepest settings screen (< 1 ms)
- Stage 7: Auto-Repair Validator — scrubs URLs, fixes word counts, orders actions safe-first (< 0.01 ms)
- Stage 8: Cache Warming — generates paraphrases, writes through to all 3 cache tiers (< 5 ms)

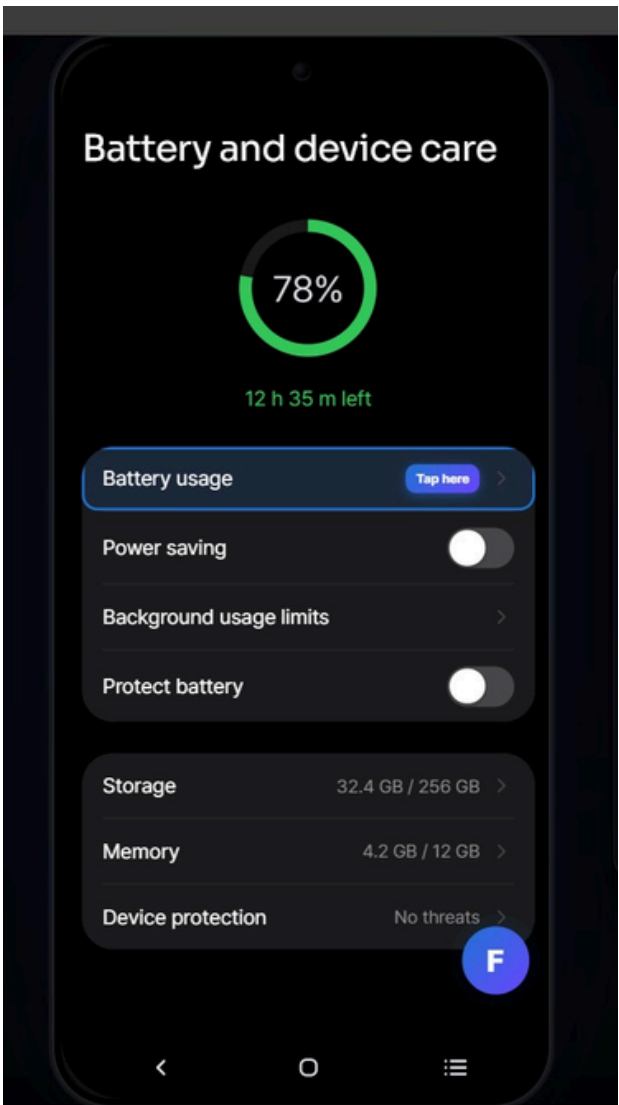


# Demo & Product Walkthrough

*From Complaint to One-Tap Fix*

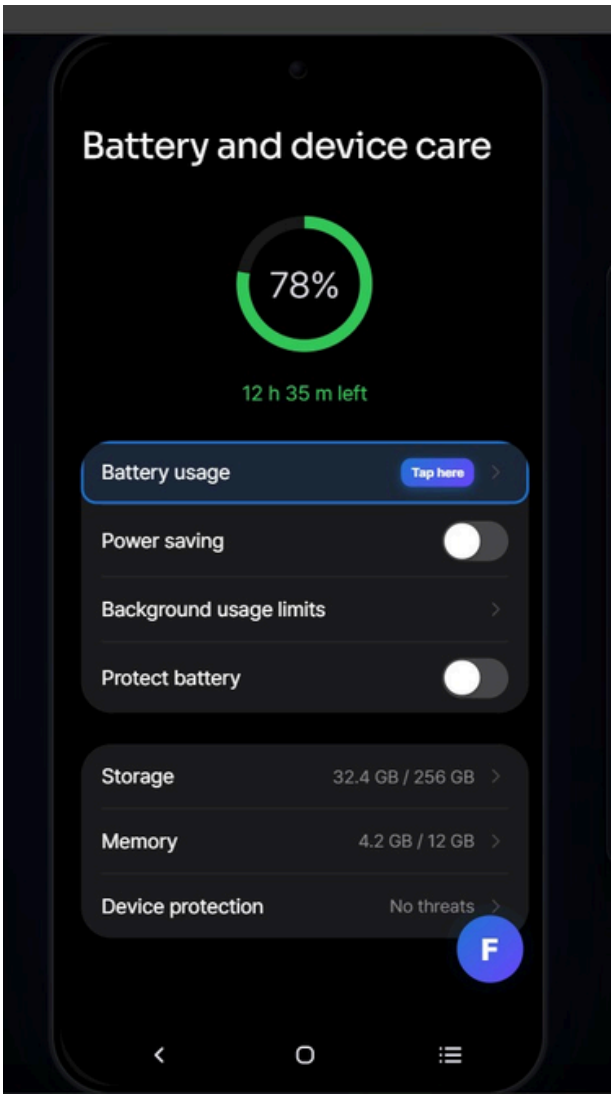
## 1. Example 1 — Cold query (first time):

- Input: "battery draining fast"
- Output: 3 structured actions with bixby:// deeplinks
- Latency: 45 ms



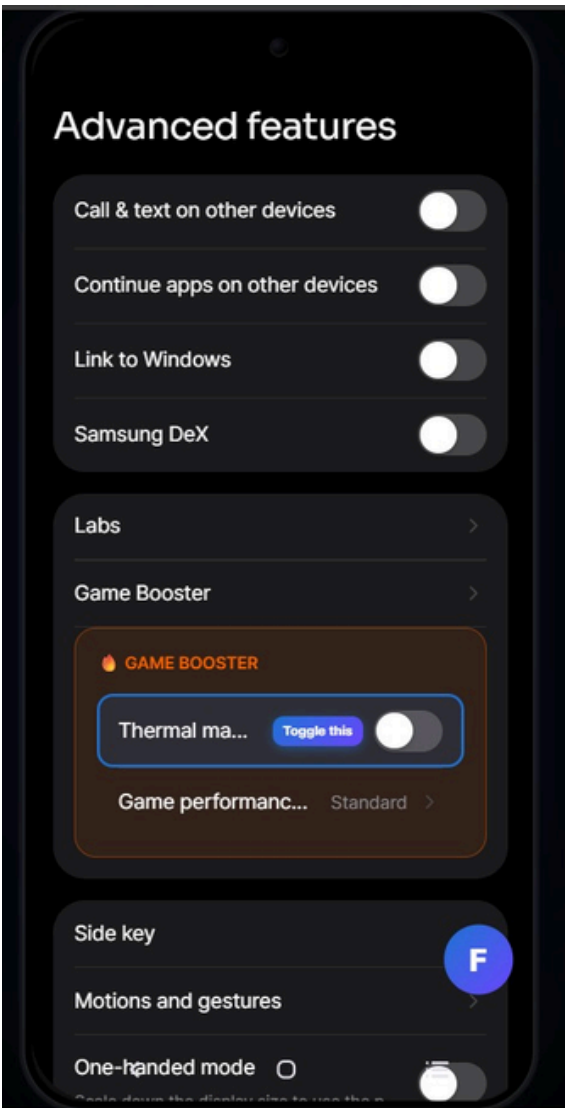
## 2. Example 2 — Same query again:

- Input: "battery draining fast"
- Output: Same response from Tier 1 cache
- Latency: 0.91 ms (50x faster)



## 3. Example 3 — Hinglish

- Input: "bhai mera phone bohot garam ho raha hai"
- Detected: domain=battery, symptom=overheating, language=hi-Latn
- Output: Tier 2 slot cache HIT
- Latency: 1.0 ms



# Tools and tech stack used

## *Production-Grade Stack*

- API: FastAPI (Python 3.11) with Swagger docs
- Schema: Pydantic v2 — Samsung-exact data contracts
- LLMs: Groq (Qwen3.8-27B) + Google Gemini 1.5 Flash — dual-provider circuit breaker with auto-failover
- Embeddings: all-MiniLM-L6-v2 (SentenceTransformer) — for cache Tier 3 and deeplink matching
- Settings Graph: NetworkX DiGraph — 112 nodes across 22 One UI categories
- Cache: Custom 3-Tier (MD5 → SHA-256 → Cosine) — no external DB needed
- Containers: Docker + Docker Compose — backend on port 8000, frontend on port 3000
- CI/CD: GitHub Actions — lint + 27 tests + benchmark on every push
- Frontend: 3D Galaxy S24 mockup, One UI simulator, diagnostic decision tree, telemetry HUD
- Telemetry: Live latency percentiles, cache hit rates, language distribution

# Impact & Use case

*15-Minute Support Calls → 1 ms Automated Resolution*

## Speed:

- Before: ~15 minutes per support ticket (human agent)
- After: 0.97 ms median response (automated)
- 70.7% of queries served from cache at zero AI cost

## Safety:

- 0 hallucinated URLs across 41 benchmark queries
- 0 safety ordering violations
- 100% schema compliance — every field validated by code

## Multilingual:

- English: 29 queries tested
- Hinglish/Hindi: 4 queries tested
- Korean: 8 queries tested

### ***For Samsung:***

*One Fixby instance behind Bixby or Samsung Members app could serve millions of daily queries at near-zero cost with guaranteed safety.*

# Innovation highlights, results and limitations

3 Innovations — Backed by Numbers

## 1. Innovation 1:

3-Tier Semantic Cache Users who say "battery draining fast" (English), "battery jaldi khatam" (Hinglish), and the Korean equivalent all hit the same cached response through different tiers. Not string matching — meaning matching.

## 2. Innovation 2:

Settings Hierarchy Knowledge Graph (SHKG) 112-node NetworkX graph across 22 Samsung One UI categories. Automatically resolves "Settings > Battery" to the deepest child: "Settings > Battery > Background usage limits > Deep sleeping apps."

## 3. Innovation 3:

Auto-Repair Validator Fixes LLM output in 0.01 ms without re-calling the LLM — enforces Samsung template, normalizes word counts, scrubs leaked URLs, orders actions safe-first.

## Benchmark Results (41 queries across 3 languages):

- Hallucinated URLs: 0
- Safety violations: 0
- Schema compliance: 100%
- Cache hit rate: 70.7% (target was 60%)
- Median latency: 0.97 ms (target was 300 ms — we are 309x faster)
- p95 latency: 1.29 ms
- p99 latency: 27.75 ms

## Limitations:

- Catalog covers 112 of 575+ screens — needs Samsung data partnership to expand
- Hinglish/Korean coverage is 12 of 41 queries — needs broader multilingual testing
- Offline fallback is a single baseline response, not full coverage

# What's next

*From API to On-Device Galaxy Intelligence*

## 1. Phase 1 — Catalog Expansion (Near-term)

- Expand from 112 to 575+ deeplinks
- Auto-generate catalog from One UI changelogs

## 2. Phase 2 — On-Device SLM (Medium-term)

- Move Stage 5 (GenAI) to a Small Language Model on Galaxy NPU
- Fully offline troubleshooting — no internet needed
- Aligns with Samsung Galaxy AI strategy

## 3. Phase 3 — Global Scale (Long-term)

- Expand to Samsung's top 10 global markets
- User feedback adjusts cache weights in real time (already implemented via /v1/feedback)
- Multi-turn escalation already works: AUTO → CAUTION → CRITICAL

# Brownie points slide( differentiation)

*What Makes This More Than a Chatbot*

## 1. GenAI is Contained, Not a Crutch

LLM is used in 1 of 8 stages. It cannot hallucinate deeplinks (Stage 4 bounds candidates), cannot violate schema (Stage 7 auto-repairs), cannot create unsafe ordering (Stage 7 partitions by code).

## 2. Dual-LLM Circuit Breaker

Groq + Gemini with automatic failover. If both fail, offline fixtures return safe responses. Engine never crashes.

## 3. Self-Auditing Benchmark

One command runs 41 queries, measures everything, writes metrics.md and results.jsonl. Every claim is machine-verified.

## 4. Multi-Turn Escalation

Tracks conversation state and attempted actions. Escalates AUTO → CAUTION → CRITICAL. Filters out already-tried fixes.

## 5. SIIS Decision Matrix

When Samsung device telemetry is available (battery %, temp, storage, signal), pipeline bypasses GenAI entirely — instant, guaranteed-correct response.

# Checklist- Updated on Public GitHub

## *Submission Checklist*

- Working prototype code — public or shared GitHub repo (Y/N) - Y
- README with reproducible setup instructions (Y/N) -  
[https://github.com/Nishant-codess/fixby\\_galaxy\\_engine/blob/develop/README.md](https://github.com/Nishant-codess/fixby_galaxy_engine/blob/develop/README.md)
- Demo video, max 5 minutes (YouTube or Drive link) -  
<https://youtu.be/CweVKNfK24I?si=kOnye3QL2gz9HHFk>
- Presentation file (PPT or PDF) (Y/N) - Yes

# Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India